

Test 14 — Question Paper

38 items. Suggested time: 95 minutes (about 2 minutes 30 seconds per item).
A four-function calculator may be used **only** on items marked *calculator*.
The Grade 6 B.E.S.T. Mathematics Reference Sheet is on the last page.
Aligned to the FAST Grade 6 Mathematics test design: 33–42% Number Sense and Operations, 25–36% Algebraic Reasoning, 25–36% Geometric Reasoning, Data Analysis, and Probability.

1. The table shows the number of minutes six students spent on the bus.

What is the mean number of minutes?

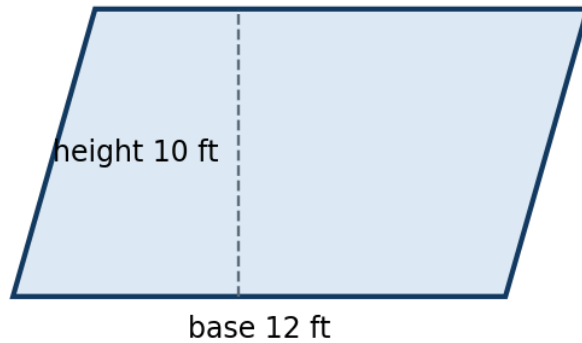
Minutes	#1	#2	#3	#4	#5	#6
Minutes	8	14	15	17	18	24

- A. 16
- B. 96
- C. 18
- D. 17

Show your work

multiple choice

2. What is the area of the parallelogram shown, in square feet?



- A. 60
- B. 44
- C. 120
- D. 22

Show your work

multiple choice

3. Which value of x makes the equation $7x + 16 = 30$ true?

- A. 2
- B. 3
- C. 1
- D. 14

Show your work

multiple choice

4. A sixth-grade class at Riviera Middle has 36 students who ride the bus and 8 students who walk.

What is the ratio of students who ride the bus to students who walk, in simplest form?

- A. 11:2
- B. 2:9
- C. 9:2
- D. 9:11

Show your work

multiple choice

5. Which fraction is equivalent to 60%, written in simplest form?

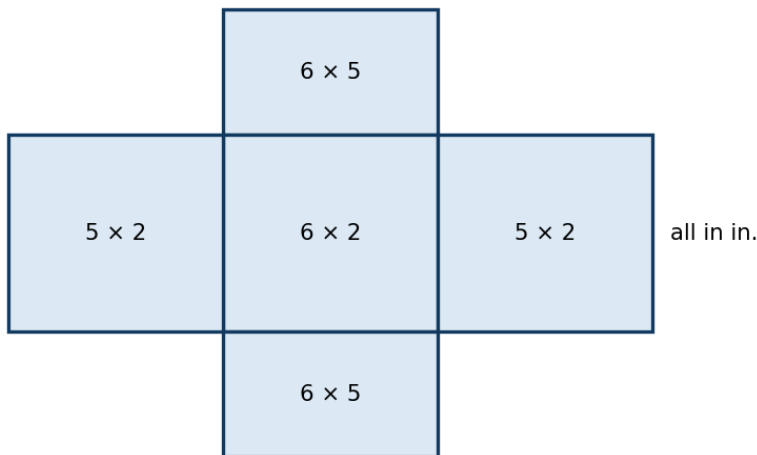
- A. $\frac{5}{3}$
- B. 6
- C. $\frac{3}{5}$
- D. $\frac{6}{5}$

Show your work

calculator · multiple choice

6. The net of a right rectangular prism is shown.

What is the surface area of the prism, in square inches?



- A. 60
- B. 52
- C. 110
- D. 104

Show your work

multiple choice

7. What is the value of $4(9 + 7)$?

- A. 64
- B. 252
- C. 20
- D. 43

Show your work

calculator · multiple choice

8. What is $2\frac{3}{5} \div \frac{1}{2}$?

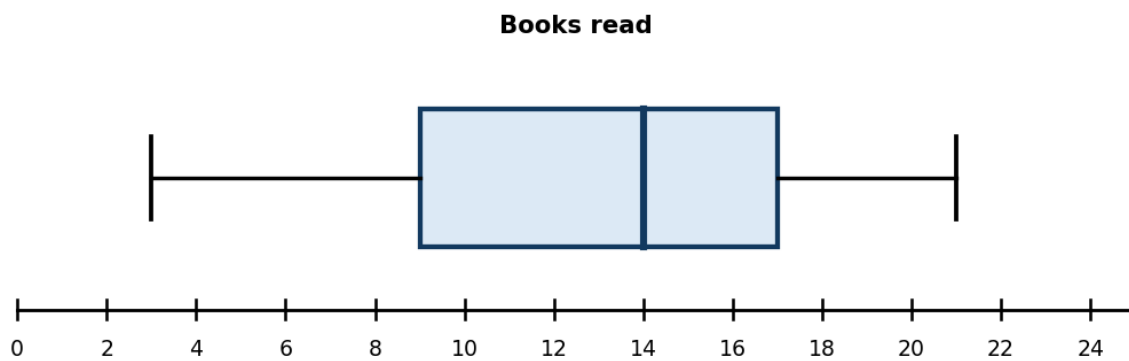
- A. $1\frac{3}{10}$
- B. $\frac{5}{26}$
- C. $3\frac{1}{10}$
- D. $5\frac{1}{5}$

Show your work

calculator · multiple choice

9. The box plot shows a set of data.

What is the median of the data?



- A. 14
- B. 9
- C. 17
- D. 18

Show your work

multiple choice

10. What value of x makes the equation $x + 5.04 = 6.74$ true?

- A. 5.04
- B. 11.78
- C. 6.74
- D. 1.7

Show your work

multiple choice

11. A pond loses 4 centimeters of water each week during a dry spell.

Which integer represents the total change in water level after 6 weeks? Enter your answer.

Answer: _____

Show your work

calculator · equation editor

12. What is the value of $|-24| - |8|$?

- A. 16
- B. 32
- C. -16
- D. 18

Show your work

calculator · multiple choice

13. A data set is shown.

2, 14, 17, 19, 21, 24, 27, 30, 37

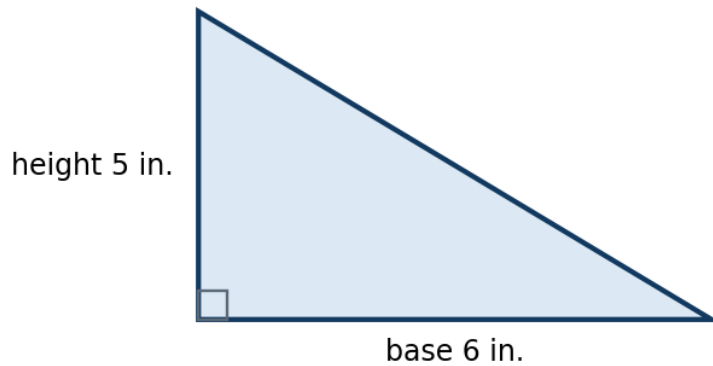
Which list gives the five-number summary (minimum, lower quartile, median, upper quartile, maximum) of the data?

- A. 2, 21, $31\frac{1}{2}$, $57\frac{1}{2}$, 37
- B. 2, 14, 21, 30, 37
- C. 2, $15\frac{1}{2}$, 21, $28\frac{1}{2}$, 37
- D. 2, $31\frac{1}{2}$, 19, $57\frac{1}{2}$, 37

Show your work

multiple choice

14. What is the area of the right triangle shown, in square inches?



- A. 15
- B. 30
- C. 11
- D. $5 \frac{1}{2}$

Show your work

multiple choice

15. There are 40 students in the sixth grade at Coral Way K-8. 40% of them ride the bus.

How many students ride the bus?

- A. 16
- B. 1
- C. 0
- D. 160

Show your work

multiple choice

16. A point is graphed at $-2\frac{3}{4}$ on a number line.

How many units is the point from 0?

- A. $-2\frac{3}{4}$
- B. $3\frac{3}{4}$
- C. $1\frac{3}{8}$
- D. $2\frac{3}{4}$

Show your work

calculator · multiple choice

17. A data set is shown.

10, 13, 21, 22, 24

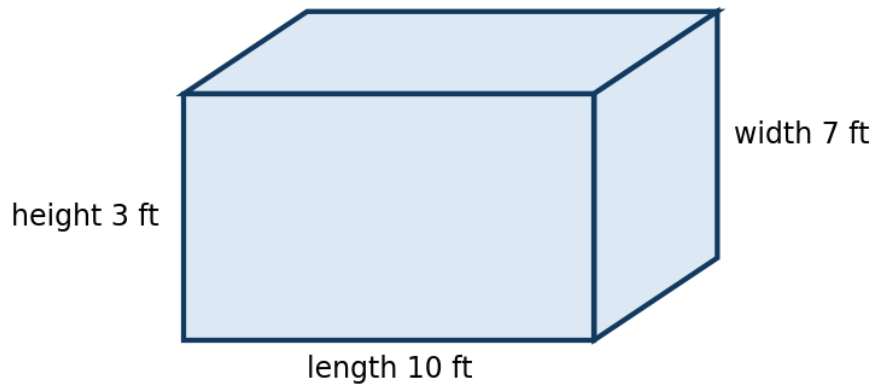
The value 25 is added to the data set. What is the mean of the new data set?

- A. 25
- B. 23
- C. 18
- D. $19\frac{1}{6}$

Show your work

multiple choice

18. What is the volume of the right rectangular prism shown, in cubic feet?



- A. 20
- B. 70
- C. 242
- D. 210

Show your work

multiple choice

19. What is the value of $5x^2 - 7$ when $x = 8$?

- A. 285
- B. 313
- C. 1593
- D. 73

Show your work

multiple choice

20. What is $2 \div 0.25$?

- A. 0.5
- B. 8
- C. 0.8
- D. 80

Show your work

calculator · multiple choice

21. Which question is a statistical question?
- A. How many text messages did each student send yesterday?
 - B. How old is my teacher?
 - C. What time does the first bell ring at my school?
 - D. How many minutes long is my bus ride to school?

Show your work

multiple choice

22. A banner is 36 feet long.

How many inches long is the banner? (1 foot = 12 inches)

- A. 108 inches
- B. 432 inches
- C. 3 inches
- D. 48 inches

Show your work

multiple choice

23. The table shows equivalent ratios of red beads to blue beads.

What number belongs in the box marked ?

Red	Blue
3	5
6	10
12	?

Equivalent ratios

- A. 9
- B. 20
- C. 12
- D. 24

Show your work

multiple choice

24. What is the value of $(\frac{1}{2})^2$?

- A. $\frac{1}{4}$
- B. $\frac{3}{8}$
- C. $\frac{1}{8}$
- D. $\frac{1}{2}$

Show your work

calculator · multiple choice

25. What is the value of $-21 - (15)$?

- A. -6
- B. -36
- C. 36
- D. -33

Show your work

calculator · multiple choice

26. Which expression is equivalent to $9(x + 6)$?

- A. $9x + 15$
- B. $9x + 6$
- C. $9x + 54$
- D. $x + 54$

Show your work

multiple choice

27. Buses on one route leave Vizcaya every 28 minutes and buses on another route leave every 42 minutes. Both leave at 9:00 a.m.

In how many minutes will they next leave at the same time?

- A. 84
- B. 1176
- C. 70
- D. 14

Show your work

calculator · multiple choice

28. Which number is located between -1 and 0 on a number line?

- A. $-\frac{3}{4}$
- B. $\frac{1}{4}$
- C. 1
- D. $-1\frac{1}{2}$

Show your work

calculator · multiple choice

29. What is the opposite of $-8\frac{1}{2}$?

- A. $-8\frac{1}{2}$
- B. $8\frac{1}{2}$
- C. $4\frac{1}{4}$
- D. -17

Show your work

calculator · multiple choice

30. A carry-on bag must weigh no more than 15 pounds.

Let x represent the weight in pounds. Which inequality represents this situation?

- A. $x > 15$
- B. $x \leq 15$
- C. $x < 15$
- D. $x \geq 15$

Show your work

multiple choice

31. Select **all** the points that are located in Quadrant III.

- (-5, -4)
- (-1, -5)
- (-3, -2)
- (4, -1)
- (2, 3)
- (-2, 6)

Select all that apply.

Show your work

multiselect

32. What is the solution to the equation $4x = 12$?

- A. 8
- B. 3
- C. 48
- D. 12

Show your work

multiple choice

33. What is the value of $-48 \div (-12)$?

- A. 4
- B. -36
- C. -4
- D. -12

Show your work

calculator · multiple choice

34. A store sells 7 pounds of guavas for \$35.00.

What is the unit price, in dollars per pound?

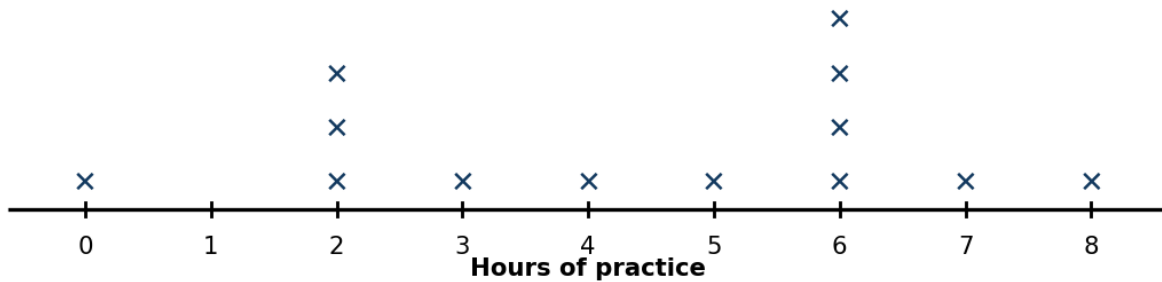
- A. \$35.00
- B. \$5.00
- C. \$1.40
- D. \$42.00

Show your work

multiple choice

35. The line plot shows hours of practice reported by a group of students.

How many students reported 5 or more?



- A. 9
- B. 7
- C. 6
- D. 8

Show your work

multiple choice

36. A number has the prime factorization $2 \times 3^2 \times 5$.

What is the number?

- A. 180
- B. 45
- C. 90
- D. 13

Show your work

calculator · multiple choice

37. Ana pays with \$100 for 3 items that cost \$5.68 each.

How much change should Ana receive?

- A. \$88.64
- B. \$82.96
- C. \$17.04
- D. \$94.32

Show your work

calculator · multiple choice

38. What is the solution to the equation $x + 9 = 46$?

- A. 46
- B. 37
- C. 55
- D. 9

Show your work

multiple choice

Grade 6 Mathematics Reference Sheet

Customary Conversions

1 foot = 12 inches	1 cup = 8 fluid ounces
1 yard = 3 feet	1 pint = 2 cups
1 mile = 5,280 feet	1 quart = 2 pints
1 mile = 1,760 yards	1 gallon = 4 quarts
1 pound = 16 ounces	1 ton = 2,000 pounds

Metric Conversions

1 meter = 100 centimeters	1 liter = 1,000 milliliters
1 meter = 1,000 millimeters	1 gram = 1,000 milligrams
1 kilometer = 1,000 meters	1 kilogram = 1,000 grams

Time Conversions

1 minute = 60 seconds	1 week = 7 days
1 hour = 60 minutes	1 year = 365 days
1 day = 24 hours	1 year = 52 weeks

Formula

Rectangular prism	$V = lwh$ or $V = Bh$
l = length, w = width, h = height	B = area of base, V = volume